Eshan Sharma

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.70
Delhi Public School, R.K.Puram	Class XII (CBSE)	2024	94.20%
Delhi Public School, R.K.Puram	Class X (CBSE)	2022	97.20%

WORK EXPERIENCE

Undergraduate Research Intern | Prof. Ankesh Jain, IIT Delhi

- Worked under academic supervision, progressing from **RTL implementation of lower-order** delta-sigma modulator loops in Verilog to the design of a **4th-order CIFB** architecture in MATLAB/Simulink.
- Contributed validated DSM implementations supporting ongoing research into high-resolution ADC architectures.

Electronics Hardware Intern | Airbotix Technology Pvt. Ltd.

- Designed **schematic** and **PCB layout** for an **LDO-based power board** using industry-standard PCB design tools.
- Assisted in schematic development for an **audio capture board**, reviewing analog signal paths, component selection, and circuit implementation while gaining exposure to mixed-signal hardware design.
- Delivered a reliable LDO board design, building knowledge of audio front-end circuit design.

Production Strategy Intern | Inter IIT Sports Meet 2025, IIT Hyderabad

- Developed operational plans and logistics strategies, and coordinated cross-functional teams with scheduling oversight.
- Coordinated cross-functional teams and managed scheduling to ensure seamless event execution under strict timelines.

PROJECTS

Solar PV Plant Performance & Anomaly Analysis (Python, Pandas, SQL)

- **Built a data-driven solar PV performance monitoring pipeline** using plant-level generation, irradiance, weather, inverter and availability data to establish expected-generation baselines and quantify performance deviations.
- **Developed plant KPIs including performance ratio, capacity factor, availability and energy losses**, separating weather-driven underperformance from equipment outages, curtailment and data-quality issues.
- **Implemented automated anomaly detection and root-cause analysis** to flag abnormal plant/inverter behavior and generate recurring performance reports for portfolio-level monitoring.

4th-Order Delta-Sigma Modulator ADC (MATLAB, Simulink, Python)

- Designed a **delta-sigma ADC** architecture for **10 MHz bandwidth**, evaluating trade-offs across modulator **order, OSR, and NTF** design parameters to meet the target resolution specification.
- Evaluated **OSR = 128 vs. OSR = 64** trade-offs, selected **CIFB over an NTF-equivalent CIFF** architecture based on internal state-swing analysis, and cross-validated the design using independent MATLAB and Simulink implementations.
- Achieved **98.46 dB SQNR** in **MATLAB** and **98.43 dB** in **Simulink** at the same operating point, corresponding to **~16-bit effective resolution**, while avoiding an impractical **2.56 GHz sampling frequency** requirement.

Design and Analysis of a 60W Open-Loop DC-DC Buck Converter (Simulink)

- Designed a **60W (24V to 12V, 5A) open-loop buck converter** in Simulink at 100 kHz switching frequency.
- Calculated **60 µH inductance and 47 µF output capacitance** to maintain <1% output-voltage ripple.
- Characterized **load and startup transients**, identifying output-voltage droop and high startup-current spikes.

Parallel FFT Engine (C++, OpenMP, Python)

- Implemented an iterative Radix-2 **Cooley-Tukey FFT**, achieving **~7,000× algorithmic speedup** over a naïve DFT implementation while validating correctness across all **62 benchmark test cases**.
- **Parallelized computations using OpenMP**, analyzed scaling behavior across multiple threads, achieved a **2.1× speedup**, and developed a **Python** automation pipeline for benchmarking, performance visualization, and result analysis.

Two-Way Set-Associative Cache Controller (SystemVerilog, Python, Icarus Verilog, GTKWave)

- **Designed and implemented an 8 KB, 2-way set-associative cache controller in SystemVerilog**, featuring 32-byte cache lines, 128 sets, write-back/write-allocate policies, 1-bit LRU replacement, and a **10-state multi-cycle FSM for hit detection, miss handling, cache-line refill, dirty write-back, and CPU response.**
- **Implemented modular tag, data, address-decoder, and LRU arrays with valid/dirty state management**, supporting 32-bit CPU accesses and 256-bit cache-line memory transfers; implemented synchronous array accesses, tag comparison, victim selection, cache-line replacement, and word-level updates within 32-byte blocks.
- **Built a Python-driven randomized verification flow generating 500 reproducible read/write transactions**, achieving **500/500 operations with 0 failures**, with GTKWave waveform analysis for FSM and memory-interface validation.

SKILLS

Programming & Data: C, C++, Python, MATLAB, SQL, Pandas, NumPy, Matplotlib

Power & Energy: Power Systems, Power Electronics, Control Systems, Solar PV Systems, BESS

Data Analytics: Time-Series Analysis, Statistical Analysis, Data Visualization, Performance Analysis, Python Automation

CMOS & VLSI: MOSFET fundamentals, CMOS logic, inverter analysis, noise margins, power/delay analysis, parasitic effects

HDL & Digital Design: Verilog HDL, SystemVerilog, FSM Design, RTL Design, Pipelining, Combinational & Sequential Logic

FPGA Development: Intel Quartus Prime, Xilinx Vivado, DE10-Lite FPGA, Virtual JTAG, TCL Scripting, Timing Analysis

DSP & Data Converters: FIR Filters, FFT, Fixed-Point Arithmetic, Quantization/SQNR Analysis, Delta-Sigma Modulation (CIFB), Noise Shaping, Oversampling, NTF Synthesis, MATLAB Verification

Tools: Simulink, LTspice, Logisim, KiCad, Git, CMake, Linux, GTKWave, Excel

RELEVANT COURSES

Digital Systems, Signal Processing, Probability and Random Processes, Machine Learning, Semiconductor Device Fundamentals, Analog Electronics, Circuits and Network Analysis, Control Systems, Engineering Electromagnetics, Power System Analysis, Electrical Machines

POSITIONS OF RESPONSIBILITY

Director | Exun Club, Delhi Public School R.K. Puram (2023-2024)

Led planning and execution of the school's annual technical fest; managed logistics and participant coordination.

Coordinator, Production | Inter IIT Sports Meet 2025, IIT Hyderabad (May '25 - Jul '25)

Managed on-ground production, scheduling, and logistics across multiple venues for a large intercollegiate event.

UG Mentor | Sunshine, IIT Hyderabad (Jul '25 - Present)

Mentored undergraduate students and supported mental-health awareness initiatives on campus.